

THE NEUTRINOS EMISSION FROM SN EXPLOSION



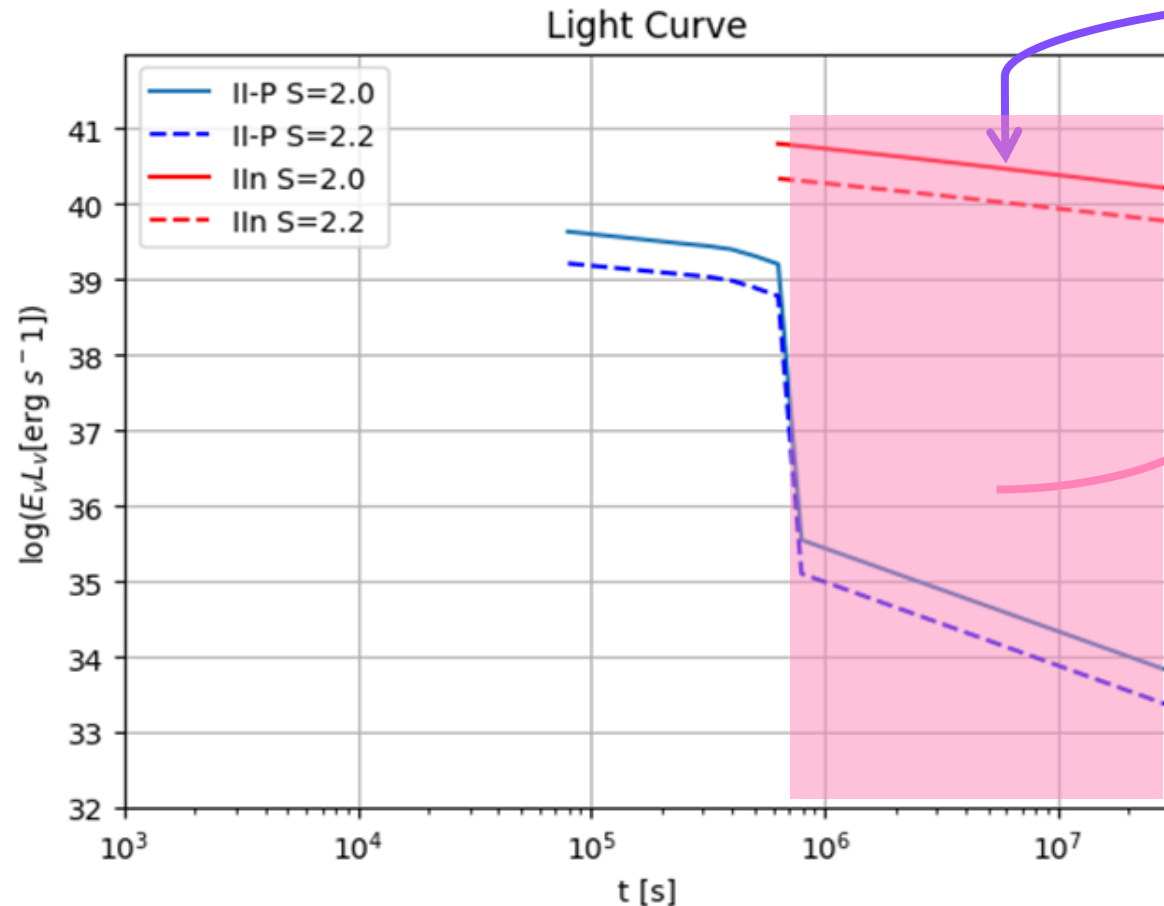
Weekly Analysis Meeting – 29/05/25

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STATUS OF THE ANALYSIS

- Cut&Count Analysis:
 - Moved to Aaet 2.6 (no bug) -> the results changed little (1-2%)
 - Done for both IIn and IIP SN model;
 - Done with ARCA8,ARCA19 and ARCA21 configuration;
 - Catalog -> on going analysis with the sources for all the configuration + combined configuration;
 - Next step: Optimization of cone & LogEtreshhold parameters.
- Unbinned Time dependent:
 - Done tests with the box function;
 - Next step: to implement the light curve from Murase in the code.

MODELING THE FLUX – SN IIN



We studied first SN IIn emission model, with $s=2.0$.

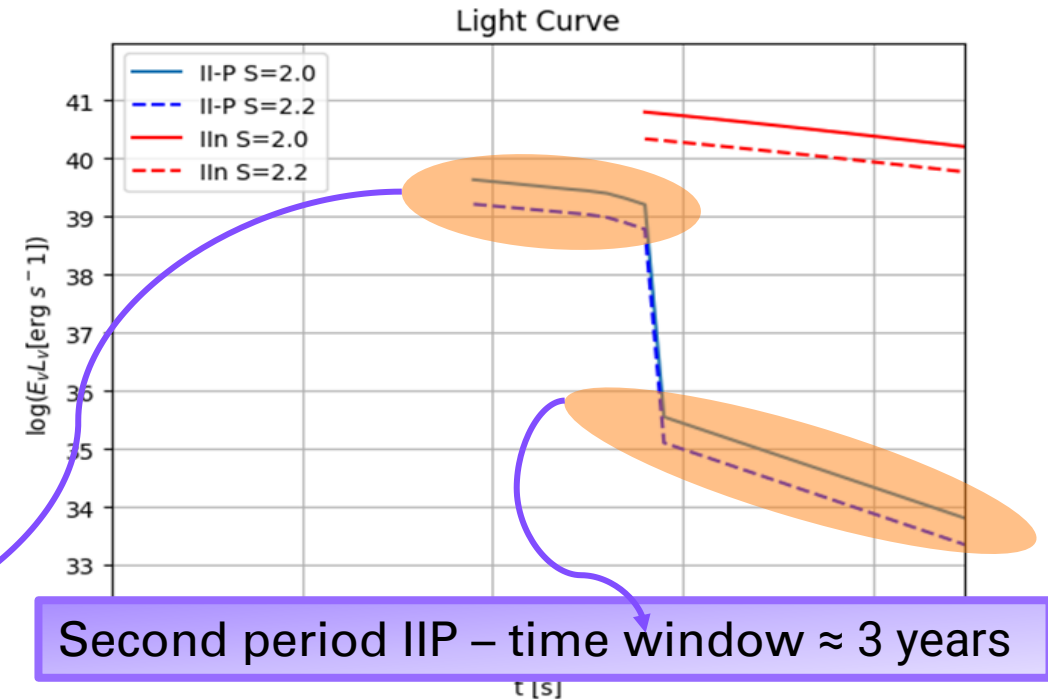
We have analyzed the time window between $t_{\text{onset}} = 10^{5.8}$ and $t_{\text{vmax}} = 10^{7.5}$

$$t_{\text{vmax}} - t_{\text{onset}} = 358,7 \text{ days} \sim 1 \text{ years}$$

MODELING THE FLUX – SN IIP

For the first time, we systematically evaluate the detectability of high-energy neutrinos from different classes of nearby SNe in a common and well-justified setup. For a Galactic SN, surprisingly, we showed that Gton neutrino detectors like IceCube and KM3Net can detect $\gtrsim 10$ neutrinos from a Type II SN, which forms a dominant class of core-collapse SNe. Even $\sim 10^3 - 10^5$ events are possible for Type II-P and IIn SNe. The targeted time window is $\sim 0.1 - 1$ day ($\sim 0.1 - 1$ yr) after the MeV neutrino detection for SNe II-P/II-L/IIb (SNe IIn). Since

First period IIP – time window ≈ 6 days



Second period IIP – time window ≈ 3 years

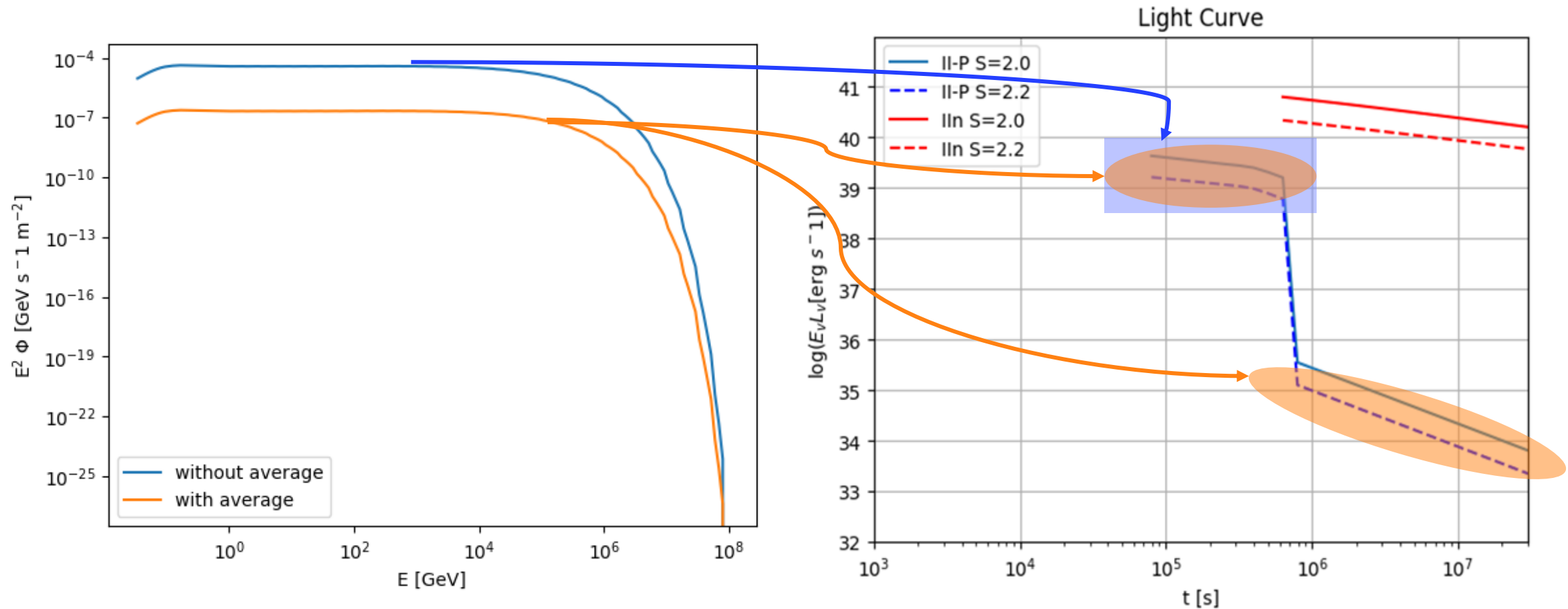
summary fortrack channel.

year	decl	cone	minlog	total-sig	total-bg	exp.limit	MRF	n	limit	discovery	MDP
0.02	-90	1	1	0.00791746	0.00523911	2.30566	291.212	1.7			214.715
0.02	-80	1	1	0.00791746	0.00523911	2.30566	291.212	1.7			214.715
0.02	-70	1	1	0.00797957	0.00556422	2.30585	288.969	1.7			213.044
0.02	-60	1	1	0.00799294	0.00591951	2.30606	288.512	1.7			212.688
0.02	-50	1	1	0.00805329	0.00630565	2.30628	286.378	1.7			211.094
0.02	-40	1	1	0.00611943	0.00484462	2.30543	376.739	1.7			277.804

summary fortrack channel.

year	decl	cone	minlog	total-sig	total-bg	exp.limit	MRF	n	limit	discovery	MDP
3	-90	1	1	8.71869e-05	0.785866	2.72204	31220.7	4.9			56201.1
3	-80	1	1	8.71869e-05	0.785866	2.72204	31220.7	4.9			56201.1
3	-70	1	1	8.76451e-05	0.834633	2.74566	31327	4.9			55907.3
3	-60	1	1	8.76336e-05	0.887926	2.77119	31622.5	4.8			54773.5
3	-50	1	1	8.76678e-05	0.945847	2.79862	31923	4.8			54752.1
3	-40	1	1	6.7278e-05	0.726694	2.69305	40028.6	4			59454.8

MODELING THE FLUX – SN IIP

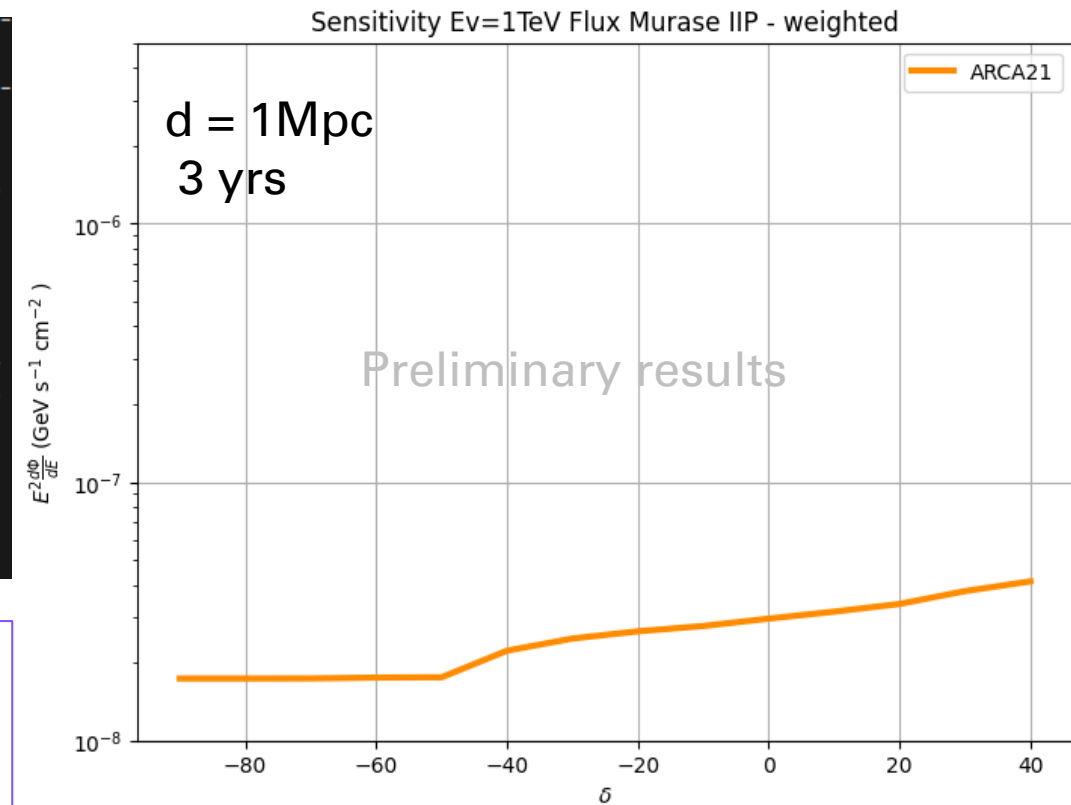


MODELING THE FLUX – SN IIP

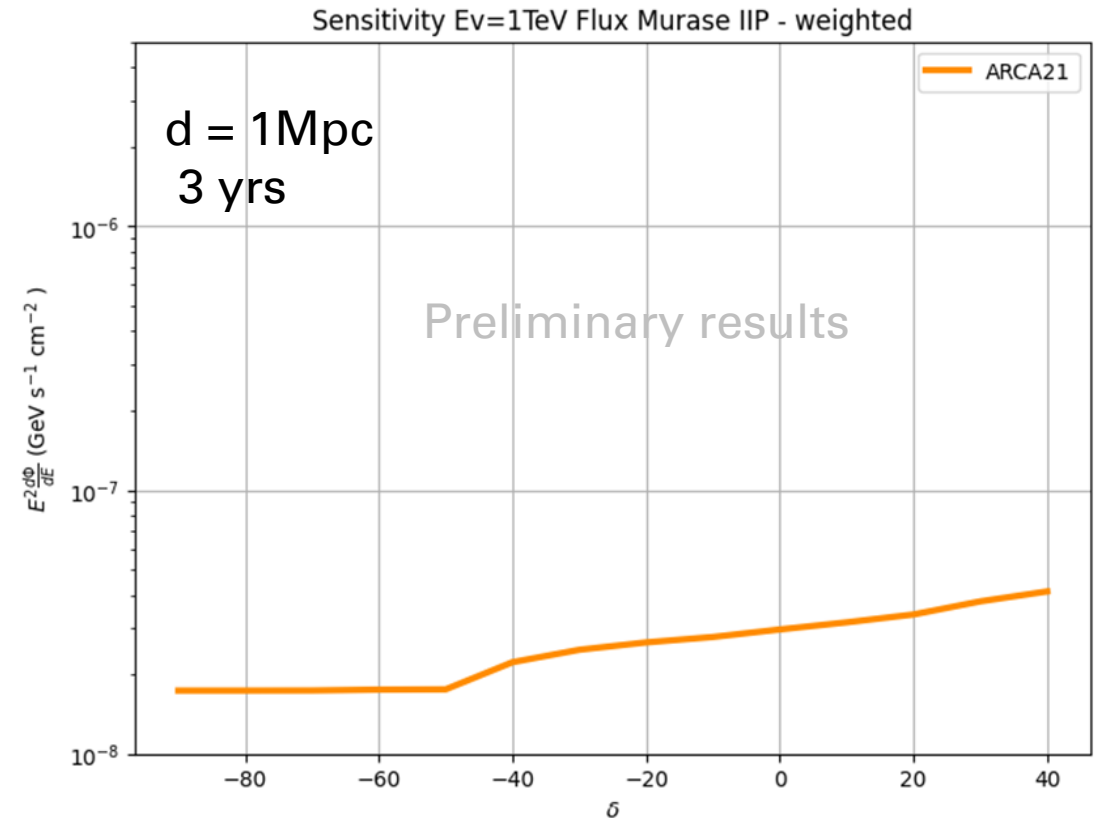
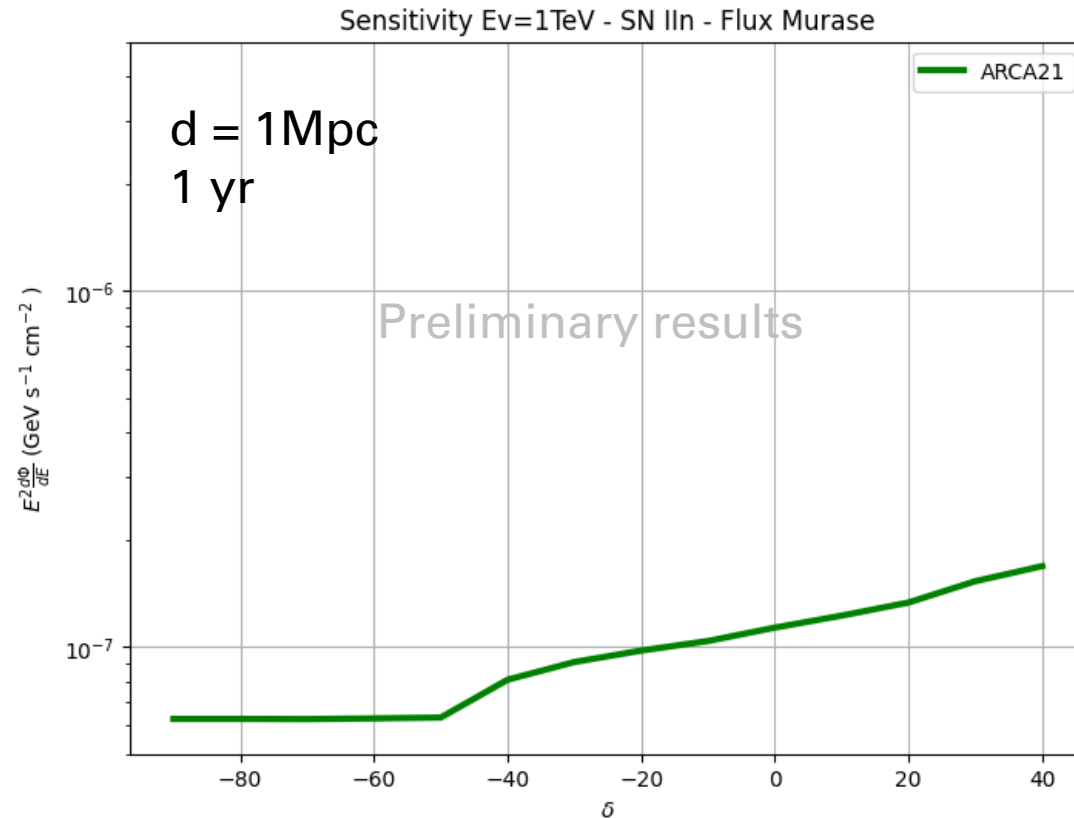
year	decl	cone	minloge	total-sig	total-bg	exp.limit	MRF	n	limit	discovery	MDP
3	-90	1	1	0.0066132	0.785858	2.72203	411.606	4.9			740.943
3	-80	1	1	0.0066132	0.785858	2.72203	411.606	4.9			740.943
3	-70	1	1	0.0066648	0.83462	2.74565	411.963	4.9			735.206
3	-60	1	1	0.00667582	0.887864	2.77116	415.104	4.8			719.013
3	-50	1	1	0.00672636	0.945732	2.79856	416.059	4.8			713.611
3	-40	1	1	0.00510612	0.726114	2.69276	527.359	4			783.373
3	-30	1	1	0.0045104	0.642161	2.65097	587.746	4.1			909.01
3	-20	1	1	0.00417973	0.579814	2.61942	626.697	4.1			980.924
3	-10	1	1	0.00394049	0.521278	2.5894	657.126	4.2			1065.86
3	0	1	1	0.0036394	0.453101	2.55391	701.741	4.3			1181.51
3	10	1	1	0.00339178	0.412366	2.53244	746.642	3.3			972.942
3	20	1	1	0.00314579	0.380709	2.51561	799.674	3.3			1049.02
3	30	1	1	0.00278347	0.344237	2.49606	896.743	3.4			1221.5
3	40	1	1	0.00254643	0.333103	2.49006	977.862	3.4			1335.2

We choose to use the average flux over 3 years ->
Results similar to the one of the time window
suggested by Murase

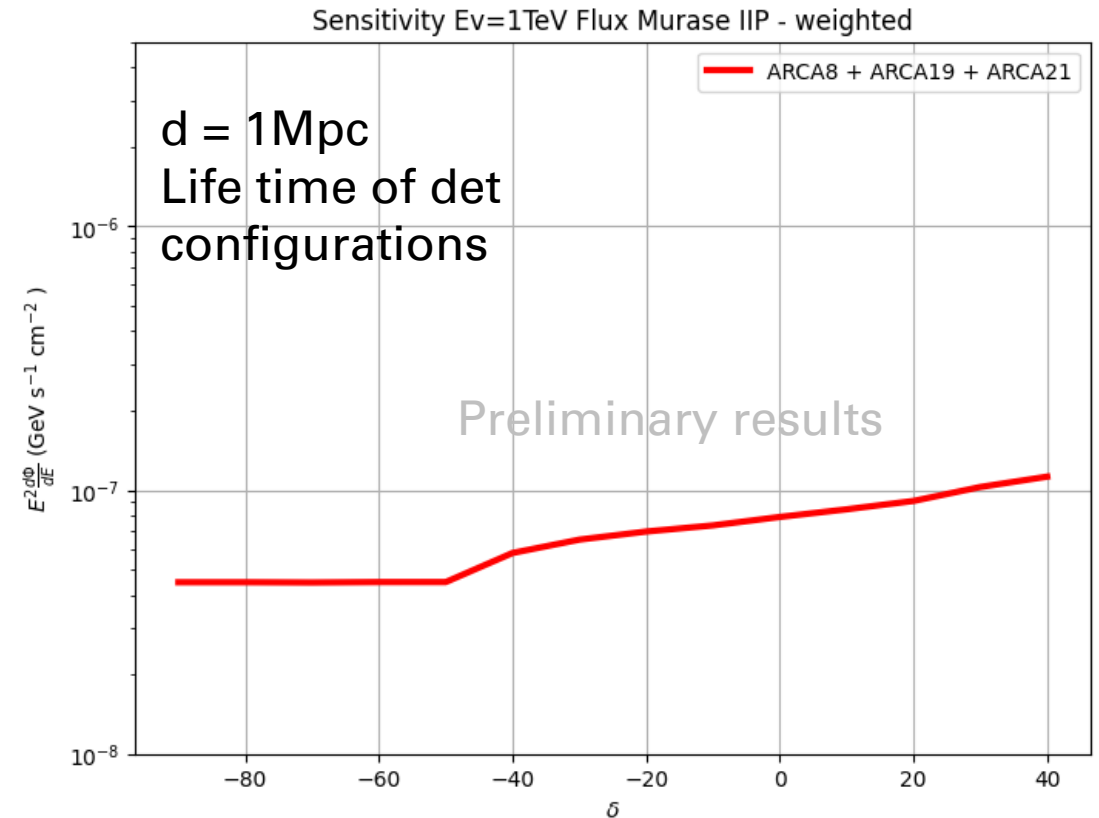
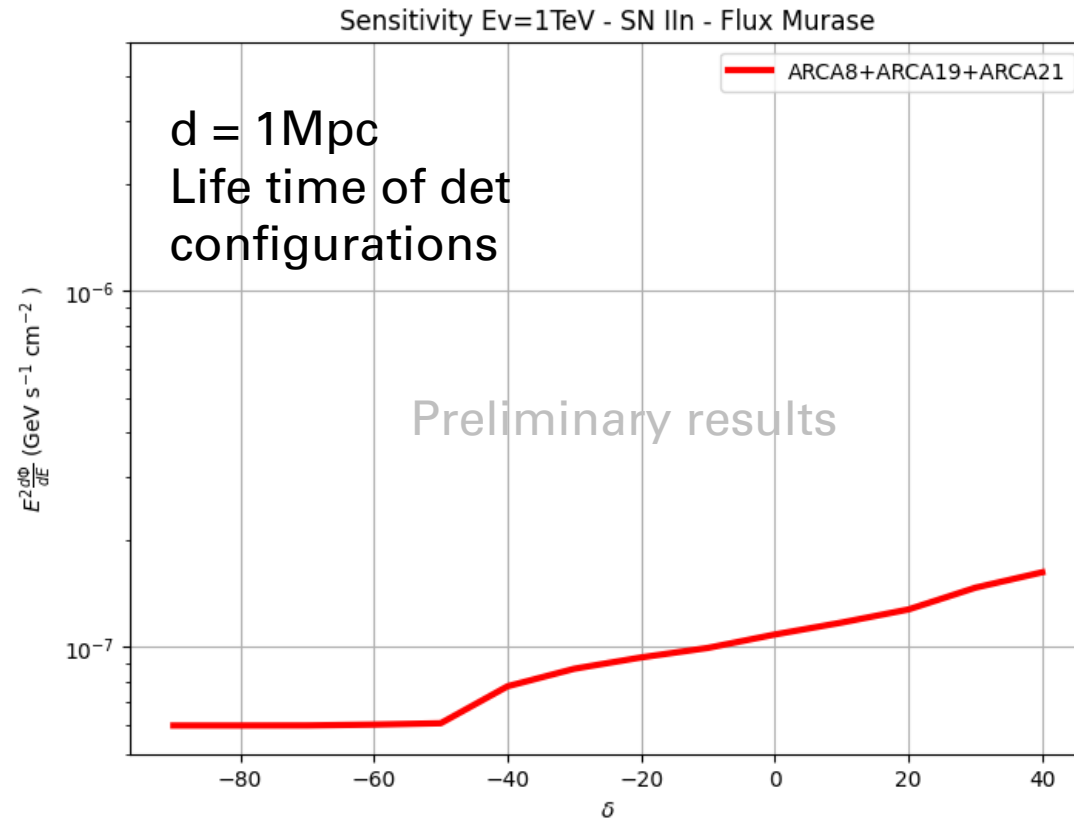
The results for the IIP will significantly improve with
the Unbinned Time dependent Likelihood!



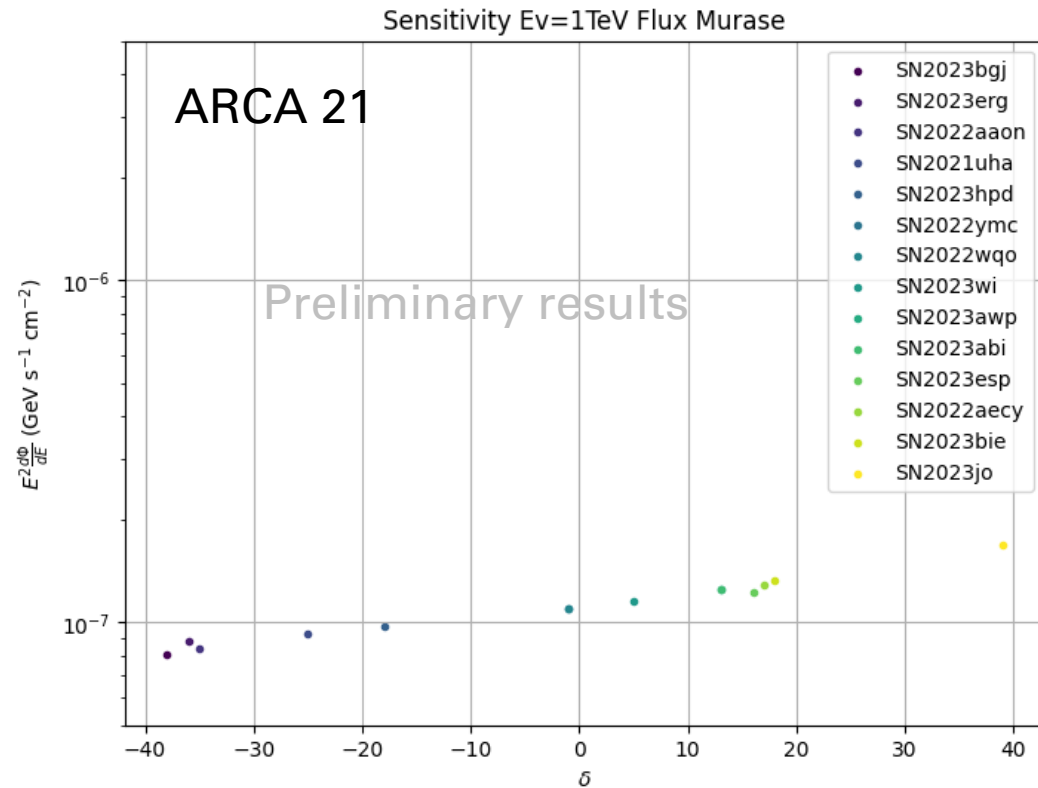
CUT&COUNT ANALYSIS - 1



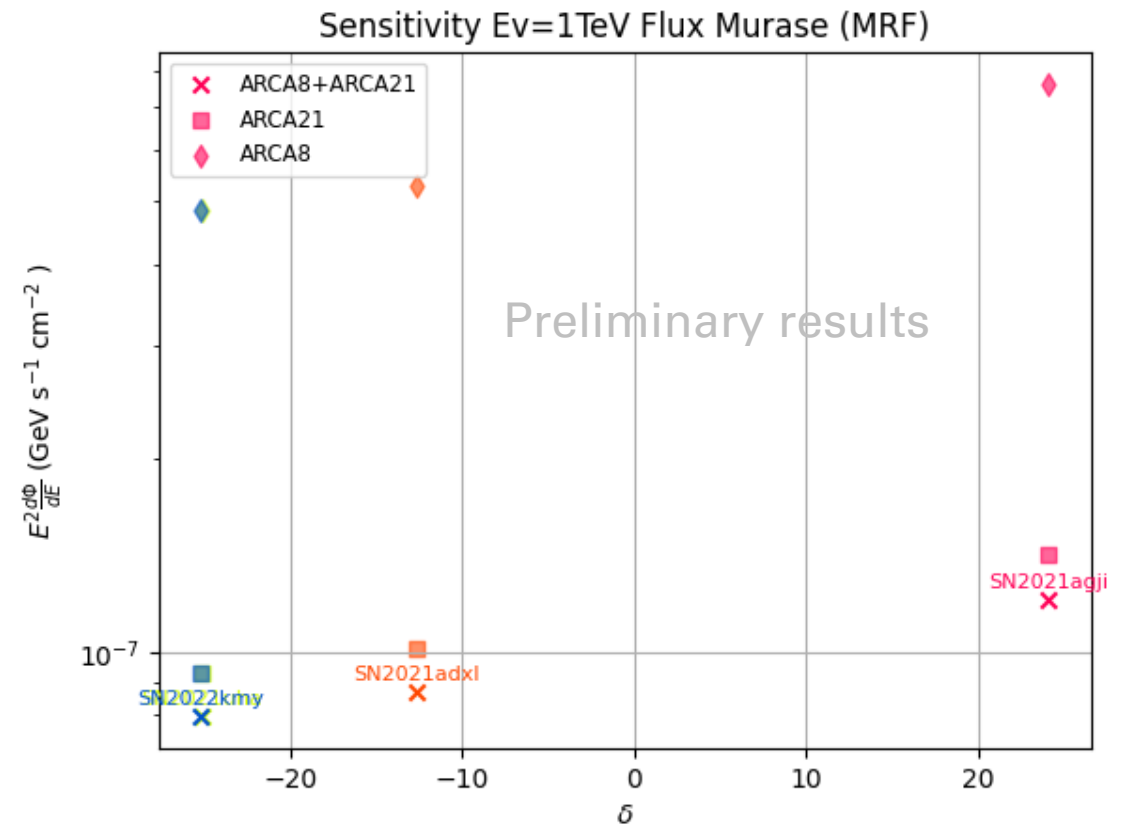
CUT&COUNT ANALYSIS - 2



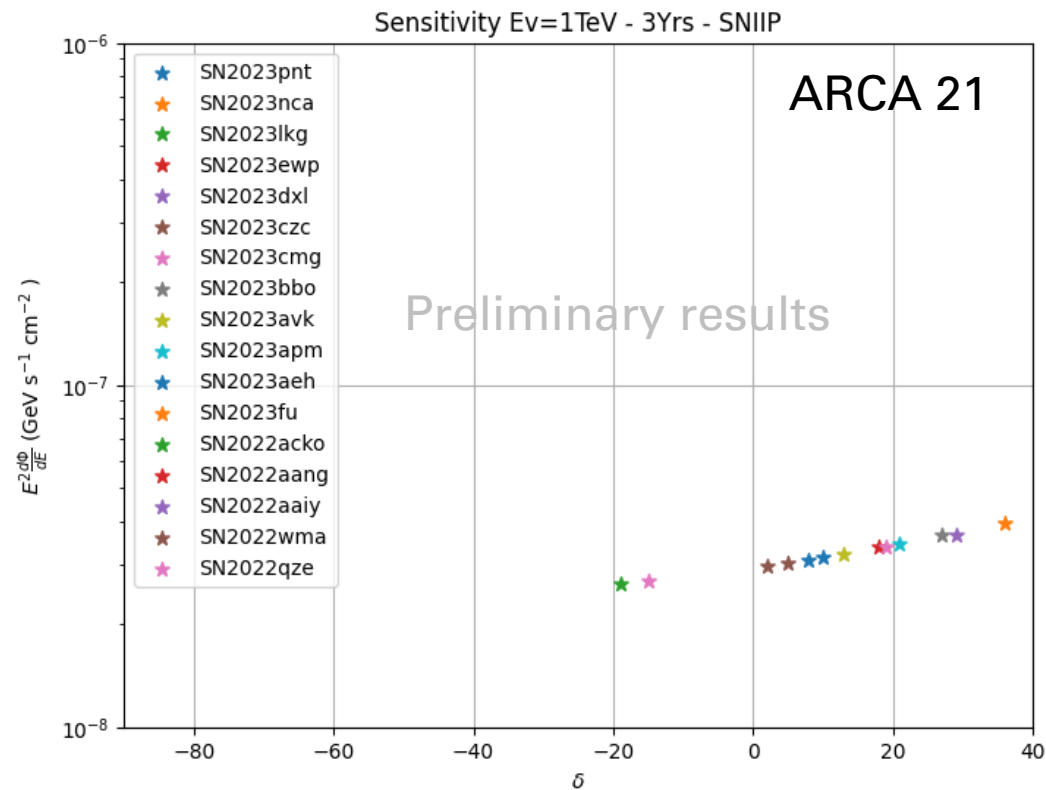
CATALOG – SN IIN



There is an overlap between different configuration for some sources

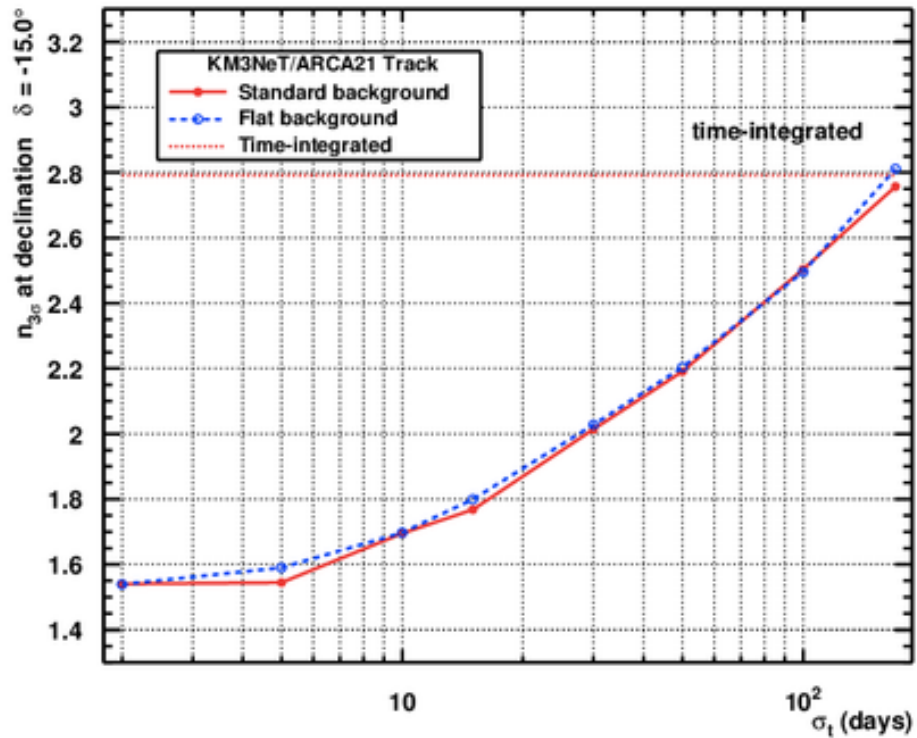


CATALOG – SN IIP



-> On going analysis with the other detector configurations

UNBINNED - TIME DEPEDENT



→ Discovery Potential from tests on box function
(Thanks to Walid, Francesco C., Meriem and Sonia)

As next step, I will help to implement
the light curve of Murase in the
framework.

$$\mathcal{T}(t) \propto (1 + t/t_{\text{PP}})^{-1}$$